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REVIEWER ACTIVITIES:
Air Force — 99
Navy — SH, MC
DLA — IS
Other — MS

USER ACTIVITIES:
Army — AV

DD Form 1426 672-1 (COORDINATED)

P.A. OS
Other Govt. AT
11

PROCUREMENT SPECIFICATION: FF-B-187

INTERNATIONAL INTEREST

SUPERSEDES:

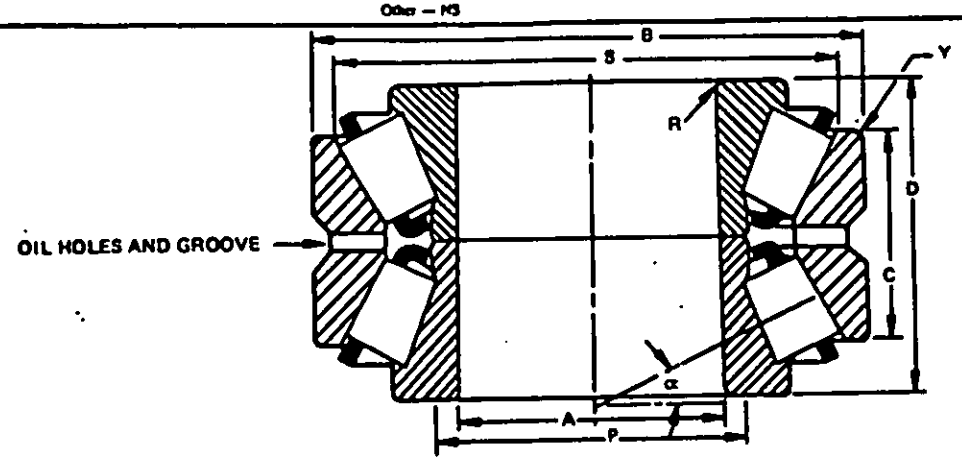
TITLE: BEARING, ROLLER, TAPERED, DOUBLE ROW OF ROLLERS, STEEP ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP, NON-ADJUSTABLE, TYPE 767 (TNA)

MILITARY STANDARD

MS 19089

SHEET 1 OF 3

3110-0520



Dash No.	AFBMA Ⓐ bearing number (See Note 11) Cone–Cup	A Bore	B Outside dia.	C Bearing width over cup	D Cones width	R Ⓐ Max. shaft fillet radius	Y Ⓐ Max. housing fillet radius	P Recommended shoulder Ⓐ diameter: Shaft	S Recommended shoulder Ⓐ diameter: Housing	Ⓐ K factor	Basic dynamic load ratings (lb) Ⓐ: Radial	Basic dynamic load ratings (lb) Ⓐ: Thrust
* 1	NA21075–21226D	0.7500	2.2500	1.4375	1.9375	0.06	0.03	1.24	2.01	0.99	3260	1730
* 2	NA43131–43319D	1.3125	3.1875	1.5625	2.1875	0.09	0.06	1.91	2.91	0.87	5500	3320
10	NA44143–44363D	1.4375	3.6250	1.5625	2.1875	0.03	0.06	2.01	3.35	0.75	6050	4250
3	NA44163–44363D	1.6250	3.6250	1.5625	2.1875	0.03	0.06	2.13	3.35	0.75	6050	4250
* 4	NA53176–53390D	1.7500	3.8750	1.7500	2.5626	0.09	0.06	2.40	3.54	0.79	7350	4900
* 5	NA55200–55444D	2.0000	4.4375	1.8125	2.5625	0.09	0.06	2.72	4.13	0.66	7900	6300
* 6	NA66212–66462D	2.1250	4.6250	2.1250	2.8750	0.14	0.03	2.87	4.37	0.93	11100	6200
11	NA72212–72488D	2.1250	4.8750	2.1875	3.0625	0.09	0.06	2.91	4.53	0.79	11900	7900
7	NA9378–9320D	3.0000	7.0000	2.9375	4.3065	0.14	0.09	4.13	6.46	0.76	22800	15800
* 8	NA98350–98789D	3.5000	7.8750	3.1581	4.5625	0.14	0.09	4.65	7.40	0.92	31500	17900
* 9	NA97450–97901D	4.5000	9.0000	3.3125	4.5625	0.14	0.09	5.51	8.38	0.79	29800	19800

*Inactive for new design after 13 Jan 82.

FSC 5110

Ⓐ denotes changes. For NOTES see sheets 2 and 3.

APPROVED 30 Apr 59 REVISED Ⓐ 13 Jan 82

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PROCUREMENT SPECIFICATION: FF-B-187

INTERNATIONAL INTEREST

SUPERSEDES

TITLE: BEARING, ROLLER, TAPERED, DOUBLE ROW OF ROLLERS, STEEP ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP, NON-ADJUSTABLE, TYPE 167 (TNA)

MILITARY STANDARD

MS 19089

SHEET 2 OF 3

3110-0520

FED. SUP. CLASS 3110

BEARINGS MANUFACTURED WITH CORRECT RUNNING CLEARANCE FOR THESE FITS.

Bearing bore A	Live shaft diameter limits	Dead shaft diameter limits		Housing	
	Tight fit	Tight fit	Loose fit	Live shaft bore limits	Dead shaft bore limits
				Loose fit	Tight fit
0.7500	0.7515–0.7510	0.7505	0.7500	2.2530–2.2520	2.2490–2.2480
1.3125	1.3140–1.3135	1.3130	1.3125	3.1905–3.1895	3.1865–3.1855
1.4375	1.4390–1.4385	1.4380	1.4375	3.6280–3.6270	3.6240–3.6230
1.6250	1.6265–1.6260	1.6255	1.6250	3.6280–3.6270	3.6240–8.6230
1.7500	1.7515–1.7510	1.7505	1.7500	3.8780–3.8770	3.8740–3.8730
2.0000	2.0020 -2.0010-	2.0005	2.0000	4.4405–4.4395	4.4365–4.4355
2.1250	2.1270–2.1260	2.1255	2.1250	4.6280–4.6270	4.6240–4.6230
2.1250	2.1270–2.1260	2.1255	2.1250	4.8780–4.8770	4.8740–4.8730
3.0000	3.0025–3.0015	3.0005	3.0000	7.0030–7.0020	6.9990–6.9980
3.5000	3.5030–3.5020	3.5010	3.5000	7.8780–7.8770	7.8740–7.8730
4.5000	4.5035–4.5025	4.5010	4.5000	9.0030–9.0020	8.9990–8.9980

NOTES:

Ⓐ 1. MATERIAL: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with FED-STD-66. Cage; carbon steel (stamped). Powdered process steel, e.g., Sinta Forge may be used.

Ⓐ 2. HARDNESS: Cones, cups and rollers: 58 to 64 Rockwell C.

3. TOLERANCES: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (A) Ⓐ			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	6.0000	10	0

Cup diameter (B)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Bearing width (C) Ⓐ			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	5.0000	100	0
5.0000	12.0000	300	0

Assembled bearing maximum radial runout		
Cup O.D. (B)		Tolerance
Over	Incl.	
0	24.0000	20

4. DIMENSIONS: All dimensions are in inches. Dimensions P and S are recommended shaft and housing shoulder diameters. Dimensions R and Y are the maximum fillet radii on the shaft and the housing respectively, which will be cleared by the bearing corners.

APPROVED 30 Apr 59 REVISED Ⓐ For changes see sheets 1, 2, and 3.

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DLA — IS
Other — NS

USER ACTIVITIES:
Army — AV

5. **OPERATING TEMPERATURE:** Recommended operating temperature not to exceed 121° Celsius (250° Fahrenheit).

6. **LUBRICATION:** Bearings shall be furnished without lubrication. Bearings shall be furnished with preservative per MIL-C-11796, Class 3.

Ⓐ 7. **BASIC DYNAMIC LOAD RATING:** Basic dynamic load rating is that constant stationary load which a group of apparently identical bearings with stationary cups (outer rings) can endure for a rating life of 90 million revolutions of the cone (inner ring). The basic dynamic load ratings listed herein are based on a rated life of 90 million revolutions or 3000 hours at 500 r/min.

To compare the load ratings on this document with others whose basis for rating are other than 90 million revolutions or 3000 hours at 500 r/min use the following formula:

$$C = \text{Other Bearing Rating} \times \left(\frac{R_1}{500}\right)^{1/f} \times \left(\frac{H_1}{3000}\right)^{1/f}$$

where:

R_1 = r/min at which other bearing is rated
H_1 = Hours life at which other bearing is rated
f = Other bearing fatigue factor

Ⓐ 8. **RATING LIFE (HOURS):** Rating life is the number of hours at some constant speed of the cone (inner ring) that 90 percent of a group of apparently identical bearings will complete or exceed before first evidence of fatigue develops. The magnitude of the rated life in hours is found from the following:

$$L_{10} = \frac{1.5 \times 10^6}{R}\left(\frac{C}{P}\right)^{10/3} \text{ hours}$$

where:

C = Basic dynamic load rating, lb.
P = Equivalent load (combined radial and thrust load), lb.
R = Revolutions per minute, r/min.

The average life is approximately four times the rating life.

9. **K FACTOR:** The K factor is the ratio of the basic radial dynamic load rating (for a one-row bearing) to basic thrust dynamic load rating.

Ⓐ 10. **CONTACT ANGLE:** All bearings are steep angle bearings having a contact angle (=) between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

Ⓐ 11. **PART NUMBER:** The MS part number consists of the MS number, plus the dash number. Example: MS19089-3. The AFBMA (Anti-Friction Bearing Manufacturers Association) cup and cone numbers are for reference only and are not to be used for ordering purposes.

Ⓐ 12. **CAGE CLEARANCE:** Designers should provide a clearance of .125 inch minimum between the outside edge of the cage and the housing counterbore.

13. For design feature purposes, this standard takes preference over procurement documents referenced herein.

14. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

APPROVED 30 Apr 59 REVISED Ⓐ For changes see sheets 1, 2, and 3.

FED. SUP. CLASS 3110

MILITARY STANDARD

MS 19089

SHEET 3 OF 3

TITLE: BEARING, ROLLER, TAPERED, DOUBLE ROW OF ROLLERS, STEEP ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP, NONADJUSTABLE, TYPE 167 (TNA)

INTERNATIONAL INTEREST

SUPERSEDES

PROCUREMENT SPECIFICATION FF-B-187

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